Pusat Sains Matematik UNIVERSITI MALAYSIA PAHANG AL-SULTAN ABDULLAH	**SUBJECT: BSD3513** **INTRODUCTOION TO ARTIFICIAL INTELLIGENCE** **TOPIC:** Chapter 2: Search Algorithms **LAB REPORT 2**	**MARKS**: 25(5%)

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Mark for CO1: /5

Rubric for CO2.

Instruction: For CO2, assess each item using the given scales.

Demonstrate critical thinking ideas of artificial intelligence knowledge in problem-solving situation.								
Item Assessed (Cognitive)	**Very Poor 0**	**Poor 1**	**Fair 2**	**Good 3**	**Very Good 4**	**Excellent 5**	**Weigh tage**	**Score**
Apply and analyse relevant artificial intelligence knowledge.	The work has not done.	Poorly applied and analysed relevant artificial intelligence knowledge and results.	Applied and analysed relevant artificial intelligence knowledge but failed to achieve successful results.	Applied and analysed relevant artificial intelligence knowledge but arrive at satisfactory results.	Applied and analysed relevant artificial intelligence knowledge to arrive at successful results.	Applied and analysed relevant artificial intelligence knowledge to arrive at excellent results.	0.5	
Using logical, rational or problem-solving appropriate to the artificial intelligence problems.	The work has not done.	The work needs to demonstrate logical, rational or problem-solving understanding appropriate to the artificial intelligence problems.	The work has demonstrated some logical, rational or problem-solving understanding appropriate to the artificial intelligence problems.	The work has demonstrated logical, rational or problem-solving understanding appropriate to artificial intelligence problems.	The work has demonstrated a thorough, logical, rational or problem-solving understanding appropriate to artificial intelligence problems.	The work has demonstrated a thorough and classy logical, rational or problem-solving understanding appropriate to artificial intelligence problems.	0.5	
						Total Score	**1**	**/5**

Rubric for CO3.

Instruction: For CO3, assess each item using the given scales.

CO3: Develop an artificial intelligence system prototype using appropriate software.							
Item Assessed (Cognitive)	**Very Poor 0**	**Poor 1**	**Fair 2**	**Good 3**	**Very Good 4**	**Excellent 5**	**Score**
Utilizing the appropriate tools / software effectively	No relevant tool used.	Tools used but did not enhance solution or information clarity.	Tools used but with limited enhancement; minimal functionality demonstrated.	Tools used appropriately to produce a functional solution with clear output.	Tools used effectively to enhance clarity, performance, and solution quality.	Tools used optimally with advanced features, clear design, and effective interaction to display the solution.	
Code functionality, clarity & structure	No code constructed.	Code incomplete or mostly non-functional; unclear and poorly structured.	Partially functional code; errors present; structure somewhat difficult to follow.	Mostly functional code with minor errors; clear structure and readable.	Fully functional and well-structured code; clearly commented and readable.	Fully functional, optimized, modular, and well-documented code; demonstrates best practices.	
Deployment & Version Control (GitHub + Streamlit or etc)	No deployment and no GitHub repository.	GitHub repo exists but incomplete OR app deploy attempt failed.	GitHub repo available with basic files; deployment page exists but app not functioning correctly.	Working deployment provided; GitHub repo contains main code files.	Working deployment with complete repository (README, code, requirements); clearly accessible.	Fully deployed app with professional GitHub repo (README, screenshots, instructions, modules, tags); live Streamlit app runs smoothly and reliably.	
						Total Score	**/15**

Question 1

Three Real-World Applications of Genetic Algorithms:

1. Automated Structural and Aerodynamic Design Optimization

Aerospace engineers parameterize complex aircraft wing geometries and use GAs to optimize lift-to-drag ratios and structural weight while satisfying stress constraints.

2. Power Grid Economic Load Dispatch (ELD)

Electric utility companies utilize GAs to allocate power generation demands among active generators to minimize total fuel consumption and emissions while respecting operational limits.

3. Financial Portfolio Optimization

Investment firms apply multi-objective GAs to construct asset portfolios that maximize expected return while minimizing investment risk (variance) over historical market data.

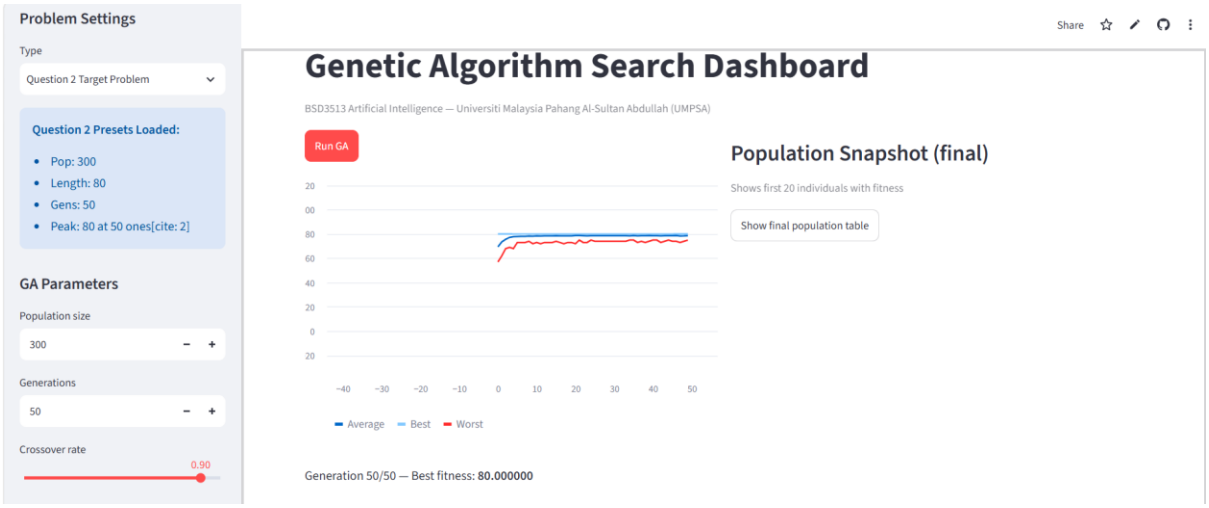
Question 2

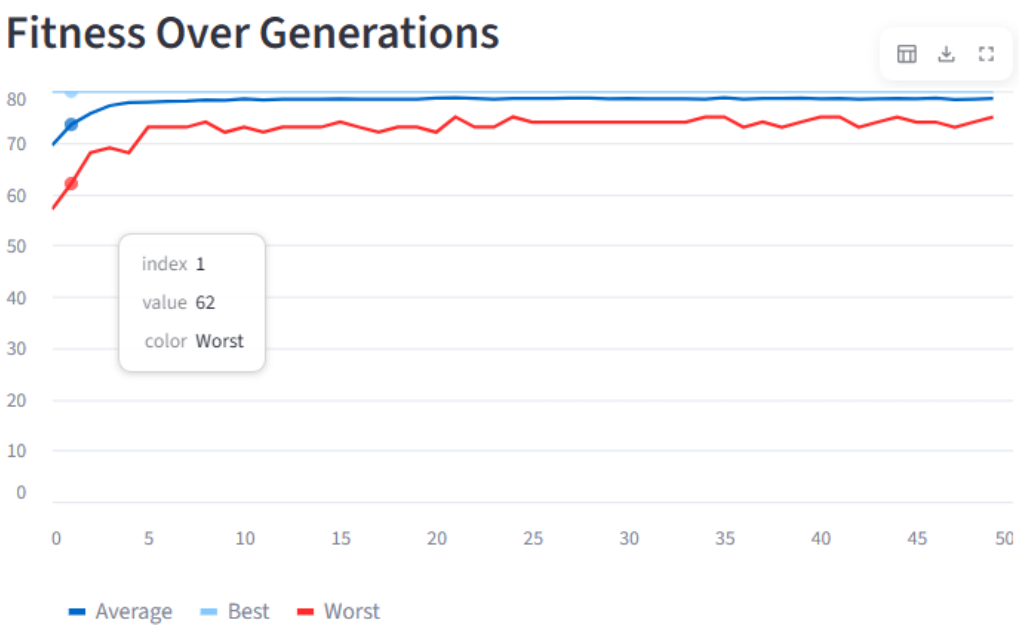
GitHub repository link

[danishAiman62/GA_lab2: Search Algorithms (Genetic Algorithm) by generate a bit pattern with predefined parameters from genetic algorithms.]

Streamlit application link and screenshot

[Genetic Algorithm - Lab 2 · Streamlit]





Best Solution

Best fitness: 80.000000

```
01101111100111110111001001010110110111100110011110011110010111001100 10
```

Number of ones: 50 / 80

The Genetic Algorithm successfully found the global optimum. The objective function was set to reach its peak value of 80 when the number of ones equals 50. Looking at the Best Solution section, the number of ones is exactly 50 / 80, which confirms that the algorithm satisfied the required condition precisely.

Based on the graph, best fitness (light blue line) quickly rises to the maximum value of 80 within the first few generations and stays flat at the top, showing rapid convergence towards the optimal bit pattern. Average & worst fitness (dark blue & red lines) start lower around 60 because the initial population is generated randomly, but steadily climb and stabilize as selection, crossover, and mutation weed out less fit individuals over the 50 generations.

The graph demonstrates the core concept of natural selection ("survival of the fittest"): the population evolves generation by generation, accumulating better genes until the overall population quality converges near the optimal solution.